



*Better study,
better decisions.*

Bannerughatta Site Report

Site 1 · Sy. 21 & 22 · 8.62 acres

SITE ANALYSIS

Prepared for Qnit case-study portfolio
Presented by GeoAce Studio Pvt Ltd

Bannerughatta, Anekal taluk, Bengaluru Urban · 12.817257° N, 77.581658° E
10 Oct 2026 / Rev A / QNIT-BLR-S01





The site at a glance

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru



LATITUDE

12.817257° N

LONGITUDE

77.581658° E

STUDY EXTENT

371 m

North up · scale bar on image
Imagery Wayback 24 Sep 2026

SOURCE / Esri World Imagery (Wayback 2026-09-24). Boundary from Case_Studies.kml, Site 1; vertices V1–V13, centroid in WGS 84.

SITE AREA

34,902 m²

8.62 acres · 3.49 ha · 345 guntas

PERIMETER

861 m

KML boundary, Oct 2026

BOUNDARY

13 edges

Irregular polygon, about 292.0 × 167.0 m

Design brief

A 3.49 ha vacant, sloping parcel off Bannerghatta Road, 456 m from Bannerghatta National Park. What can it carry under agricultural zoning and the park's eco-sensitive rules, and where do the slope, the park edge and the pending mutation set the limit?

Primary opportunity

A park-edge setting 123 m from SH-87, with a bus stop at 128 m and four schools within a 10-min walk. That suits a low-density, landscape-led use: institutional, nature education or farm-house living.

Key constraint

97.6% Agriculture on the draft RMP 2031 layer. The park boundary lies 456 m W, so the eco-sensitive zone (100 m–1 km, under Supreme Court review) may apply. Sy 22 also has a disputed mutation pending.

REPORT ROUTE

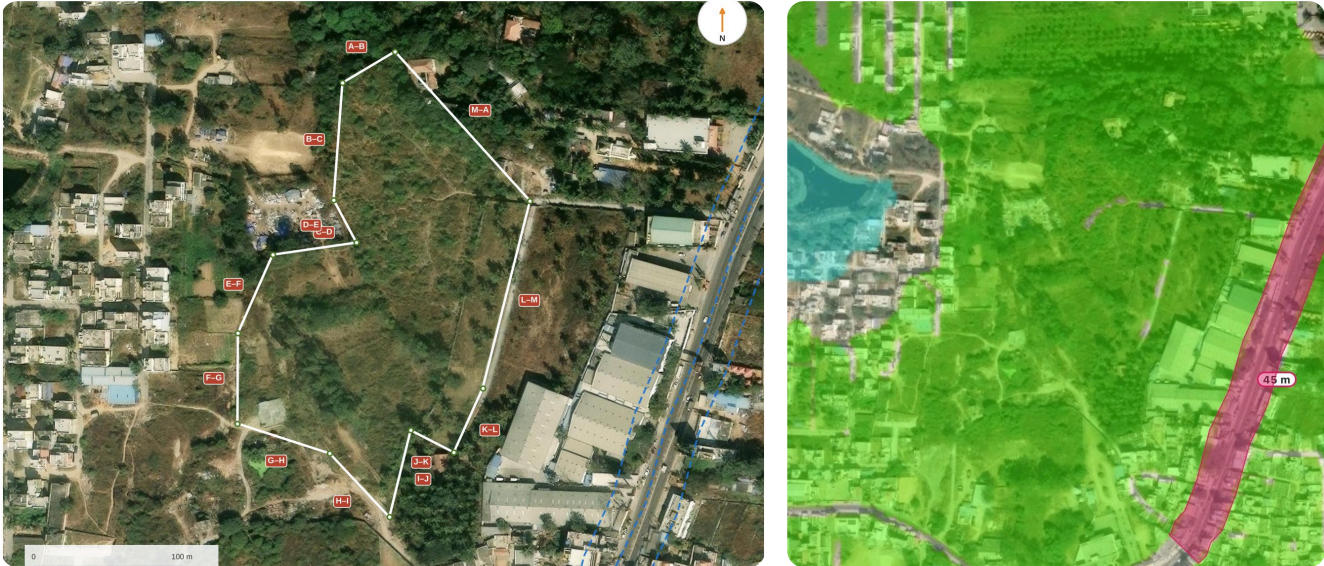
03–04 Site & ground 05–08 Environment
09–11 Context & services 12 Priorities & sources

Stakeholders: Qnit case-study portfolio / GeoAce Studio design team / Reviewer: pending



Boundary, edges & development

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru



SOURCE / Left: Esri imagery 2026; dashed = draft RMP 2031 plan roads (Qnit planning layer). Right: Qnit 2031 plan layer, BDA RMP 2031 (draft) on satellite, supplied screenshot.

Site identity

Survey / parcel **Sy. 22/*/1 (in) · 21/*/* (63 m out)**
Village / authority **Bannerughatta (613132) · BDA**
CRS / boundary **WGS 84 KML · UTM 43N**
Records / status **4 RCCMS disposed · MR 2/2026-27 pending**

Development controls

Permitted use / source **Agricultural · RMP 2015 (PD 30)**
FAR / coverage **Farm house ≤ 250 m² plinth**
Setbacks / plot **Plot ≤ 1.2 ha per farm house**
Height limit / basis **G+1 · ESZ rules apply**

Edge conditions

Edges	Adjoining condition	Clearance / width	Evidence
A–G · W (330 m)	Bannerughatta village: 1–2 storey houses, a waste and debris yard near D–E; lake beyond	Houses adjoin; lake 134 m W; park 456 m W	Esri 2026 · Overture
G–J · S (157 m)	Scrub and earth tracks; village lanes and houses to the south	No frontage road; housing 42 m S	Esri 2026 · Overture
J–M · E (211 m)	Earth track; industrial sheds, then Bannerghatta Road (SH-87)	Industrial 21 m SSE; 45 m plan road 104 m ESE	Overture · RMP 2031 layer
M–A · NE (136 m)	Houses and a track; Ryan International School north	School 80 m N; quarry 361 m N	Overture land use

DESIGN / DEVELOPMENT IMPLICATION

Two rules, not an edge, control: Agricultural zoning (non-farm use needs DC conversion) and the park's eco-sensitive zone. Get the BDA zonal certificate and the ESZ overlay for Sy 21–22, and reconcile the KML with the survey sketch, since the Sy 21 point falls 63 m outside it.



Terrain & the ground beneath

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru



SOURCE / Copernicus GLO-30 DSM (30 m, resampled to 10 m), 2 m contours, over Esri imagery. A–B follows the fitted fall line.

ELEVATION RANGE

931 / 948 m

EGM2008 · DSM includes canopy

TYPICAL SLOPE

6.8 %

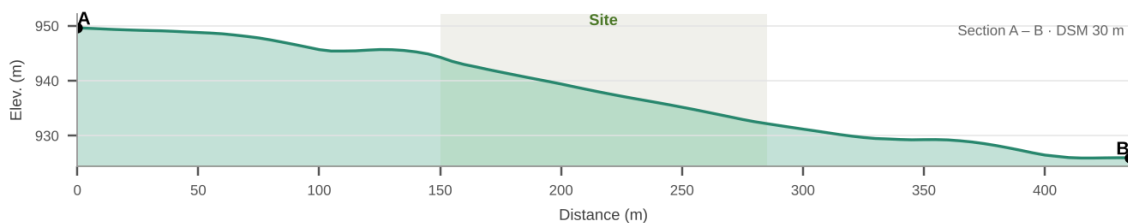
Moderate · falls to the NW

CONTOUR INTERVAL

2 m

GLO-30 DSM / 30 m

Transect A–B



SOURCE / Section A–B sampled every 5 m from the DSM; A is south-east, B north-west.

Section length 436 m / Relief 23.7 m / Vertical exaggeration ≈ 2×

DESIGN / DEVELOPMENT IMPLICATION

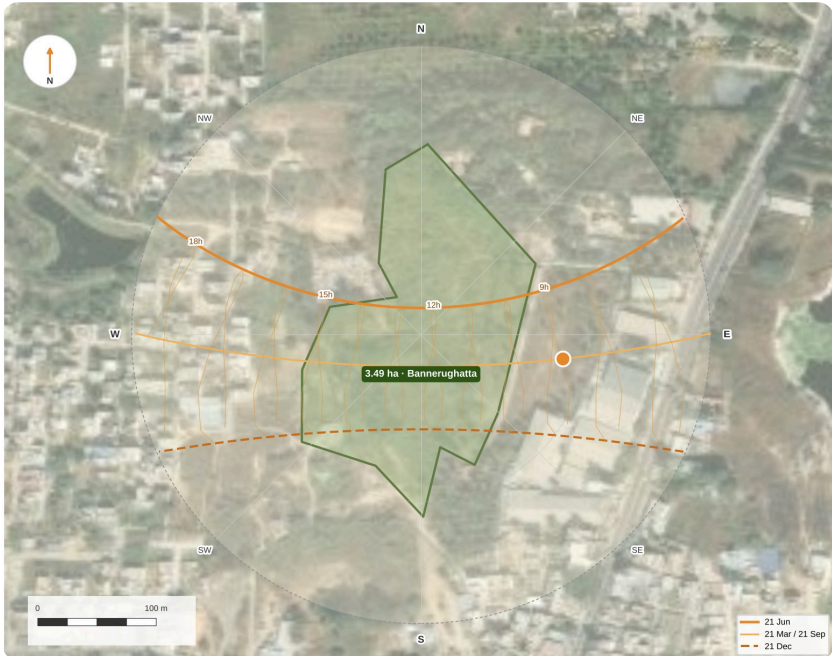
About 17 m of fall across the parcel, 6.8% to the NW; only 9% of the ground is under 5%. Terrace buildings along the contours, with the low NW corner (931 m) kept for drainage. Expect retaining of 1–2 m per terrace; confirm with a ground survey.



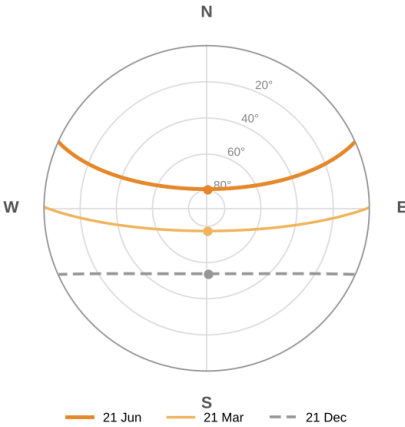
Light, shade & orientation

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru

01 / SUN PATH



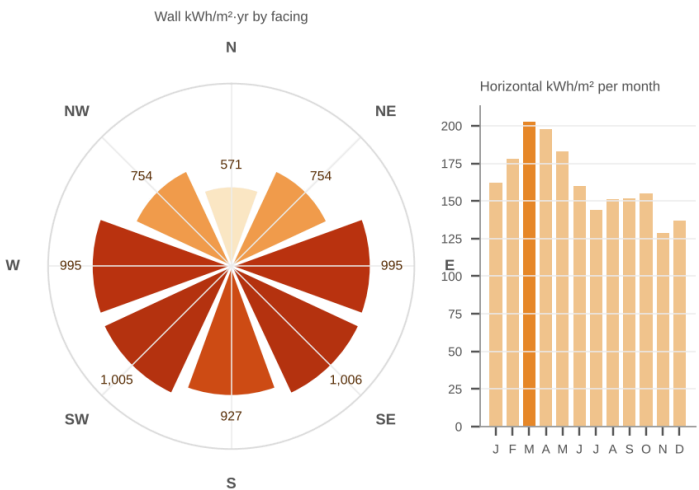
SEASONAL COMPARISON



21 Mar · 21 Jun · 21 Dec; IST (UTC+5:30)
True north; noon altitude 54° (Dec) to 79° (Jun, sun to the north)
Day length 11.4–12.9 h; no context model

SOURCE / Solar positions computed (NREL SPA, pvlib) for the site centroid; stereographic dome, Esri imagery. Hour lines at the 21st of each month.

02 / SOLAR RADIATION



ANALYSIS PERIOD

2015–2024 monthly means

INCIDENT SOLAR ENERGY

1,952 kWh/m²

Horizontal plane, per year (5.35 kWh/m²·day)
Highest wall exposure: SE / SW ≈ 1,006; north 571
Best month Mar; cloud cuts 19% vs clear sky

SOURCE / NASA POWER CERES SYN1deg monthly GHI 2015–2024; wall split modelled from clear-sky geometry scaled to it.

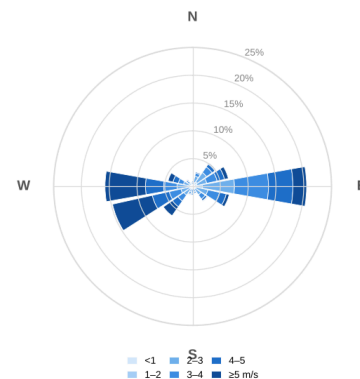
DESIGN / DEVELOPMENT IMPLICATION

The site falls to the NW, so its slope faces the cooler afternoon side. Lay terraces E–W where possible, with main rooms facing N and S: the noon sun stays above 54°, so short overhangs shade both. West walls take ≈ 995 kWh/m²·yr: shade them with the existing tree clusters.



Air movement & outdoor comfort

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru



SEASON / HEIGHT

2015–24 / 10 m

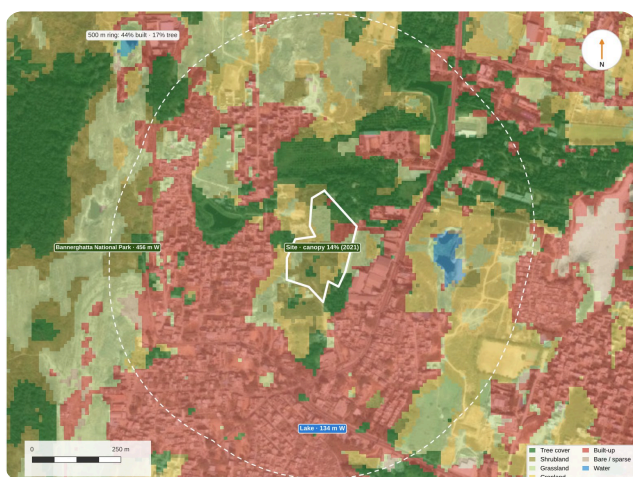
Annual: from E, mean 3.5 m/s, calm 5.3%

Jun–Sep: W 35%, 4.5 m/s

Dec–Feb: E 45%, 2.9 m/s

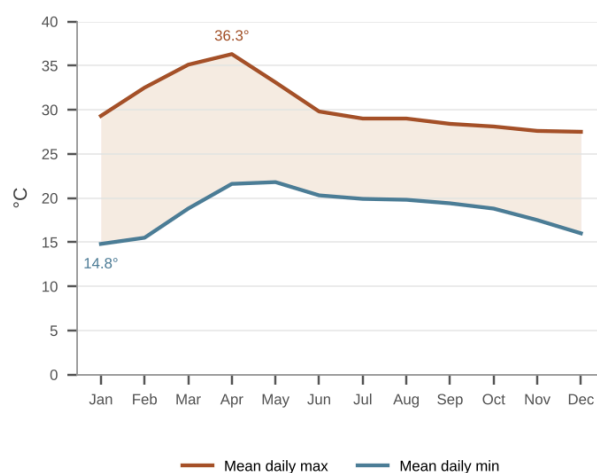
SOURCE / NOAA ISD hourly observations, HAL Airport (VOBG), 17.4 km NNE; consistent with NASA POWER MERRA-2 (from W, 3.9 m/s).

Local comfort



SOURCE / ESA WorldCover 2021 (10 m) on Esri imagery; dashed = 500 m ring.

Seasonal temperature



SOURCE / NASA POWER MERRA-2 daily, 2015–2024. HAL station mean 25.1 °C; hottest day 38.0 °C (2024-04-05).

DESIGN / DEVELOPMENT IMPLICATION

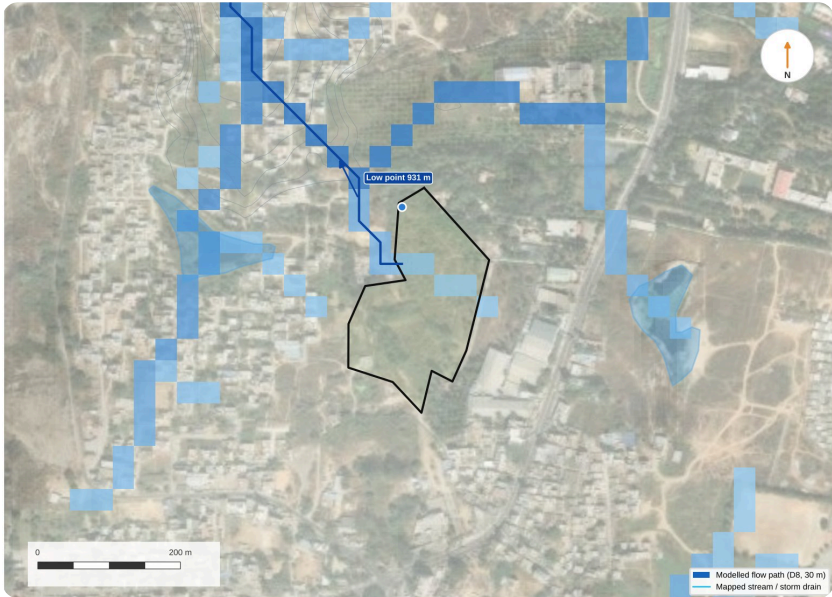
The monsoon westerly (4.5 m/s) arrives over the village and park edge; open the built form to the W and let it run down the NW slope. Winter easterlies come off SH-87 and the industrial sheds, so plant a dust and noise screen on the E edge. April is the hot month (mean max 36.3 °C).



Runoff, storage & groundwater

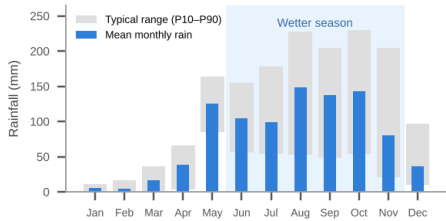
Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru

SURFACE RESPONSE



SOURCE / D8 flow on GLO-30 DSM; Overture water. Rain: NASA POWER MERRA-2 grid 13.0 N 77.5 E (shared with north Bengaluru).

MONTHLY RAINFALL

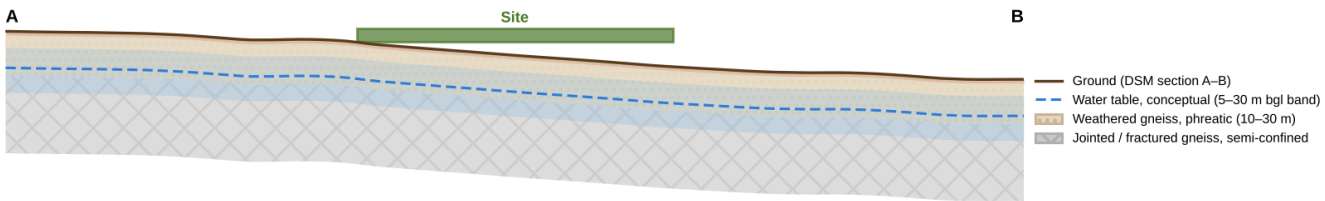


EVENT BASIS

95 mm / 24 h

10-year return period (Gumbel on 1981–2025 annual maxima)
Mean annual 937 mm; 65% falls May–Nov

Groundwater & recharge



SOURCE / Conceptual section on A–B. Strata and water depths from the CGWB Bengaluru Urban profile, not site boreholes.

Surface water

Low point NW corner (931 m) → village drain → lake 134 m W (0.8 ha).
10-yr day on 3.49 ha: $\approx 1,200 \text{ m}^3$ runoff now (C 0.35), $\approx 2,500 \text{ m}^3$ built (C 0.75).
Basis: catchment = site; no city drain data this far south.

Ground conditions

District water level 5–30 m bgl pre-monsoon (CGWB).
Anekal extraction 106% of recharge (2025-26): over-exploited.
Granite-gneiss, weathered 10–30 m; check for rock outcrops.

DESIGN / DEVELOPMENT IMPLICATION

Slope speeds runoff to the village on the NW: hold $\approx 1,300 \text{ m}^3$ in terrace swales and a detention pond at the NW corner before discharge. Groundwater is over-drawn: plan on rainwater harvesting and reuse, not new borewells.



Living systems & views

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru



SOURCE / Esri imagery; park boundary and water from Overture Maps (OSM-derived); plan-view sight lines from the site centre.

Landscape structure



Canopy cover **14% site · 17% 500 m ring**
Permeable ground **91% site · 55% ring**
Water / habitat edge **park 456 m · lake 134 m**
Tree survey / season **none · WorldCover 2021**

View & habitat priorities

- Retain**
The tree clusters on the slope and a green link west toward the park (209 km² mapped).
- Investigate**
Elephant and leopard movement at the park edge, and quarry blasting and dust (4 quarries within 1 km, nearest 361 m).

Observer: site centre, plan view (no eye-height model)
Park boundary as mapped in OSM; ESZ limits not mapped here

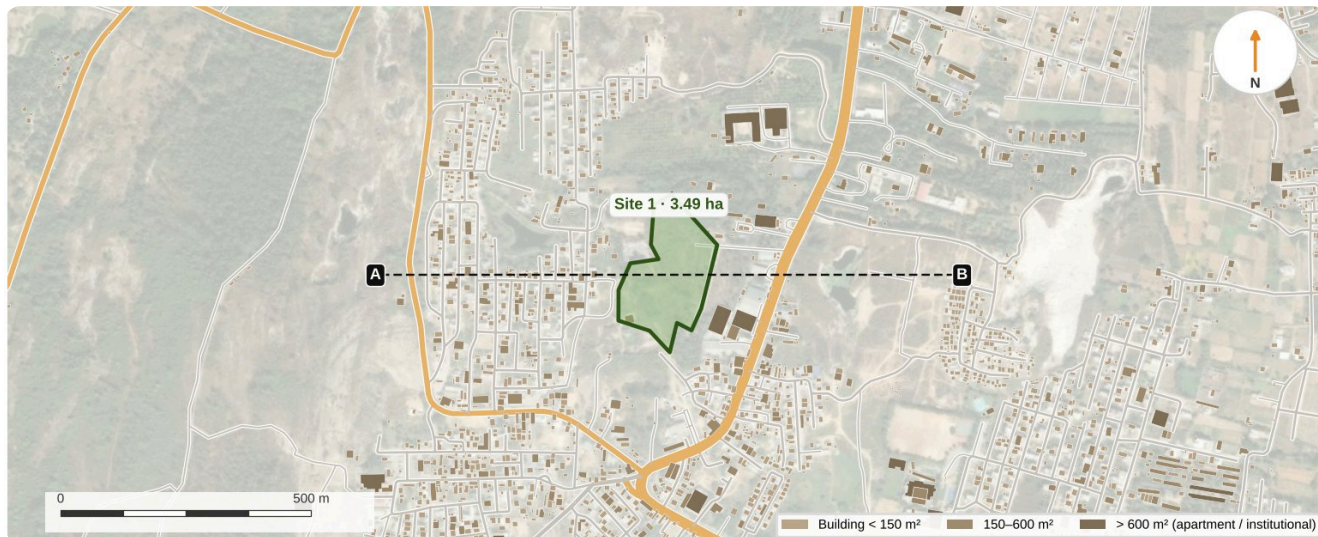
DESIGN / DEVELOPMENT IMPLICATION

Treat the west half as the landscape buffer facing the park: low, dispersed buildings, native planting, dark-sky lighting and wildlife-safe fencing. Keep built form toward the east, where the industrial edge already is.



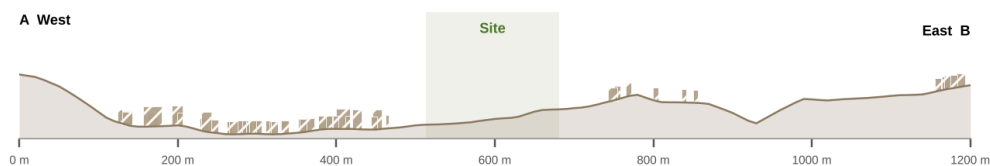
Neighbourhood fit & movement

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru



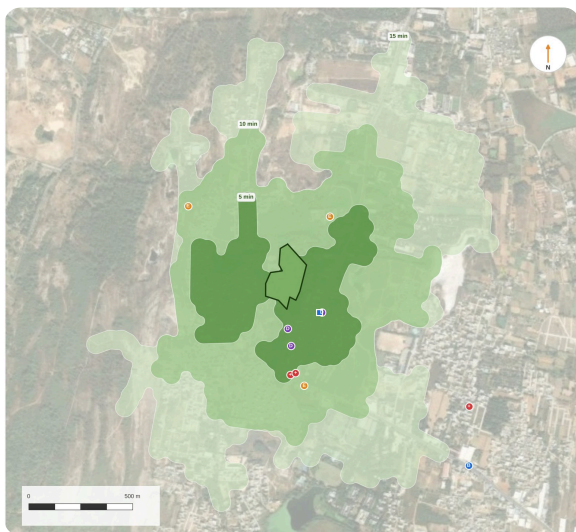
SOURCE / Overture Maps buildings and roads (release 2026-09-23) over Esri imagery; A-B is a 1.2 km W-E section through the site.

STREET & SKYLINE PROFILE A-B



54 buildings cut; none carry mapped heights, so hatched blocks use storeys assumed from footprint (2-4 floors × 3.2 m).

DAILY ACCESS



Walking 5 / 10 / 15 min at 4.8 km/h on the Overture road network from the nearest road nodes.

ACCESS BASIS

SH-87 side / walk / weekday

Transit **Vijaya Bank Bannerughatta stop** · 128 m

Bus stand **Bannerghatta BMTC** · 1.7 km

Metro **Kalena Agrahara (Pink Line)** ≈ 6 km

Health **RVM Foundation Hospital** · 7 min

Education **Ryan International School** · 7 min

Banking **Punjab National Bank** · 2 min

No public road touches the boundary; access is by earth tracks to SH-87 (123 m E). Secure a frontage or easement first.

Context: village houses (median footprint 60 m²) and industrial sheds; low-rise G+1 fits.

Overture 2026-09-23.0 · checked 10 Oct 2026



The context beyond the boundary

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru

LAND-USE CHANGE



SOURCE / Esri Wayback releases 27 Feb 2017 (left) and 24 Sep 2026 (right).

Recorded change

2017 → 2026: the parcel stays open scrub; the village grew to the W and sheds filled the E edge.

WorldCover 2021: 60% shrub and grass, 9% built.

Heritage: Champakadhama temple (Hoysala, 1257 CE) in Bannerghatta; quarries nearby.

PEOPLE & ACTIVITY



SOURCE / Built-footprint density from Overture buildings, 2026; activity proxy.

Community needs

Bannerughatta GP: about 4,062 households (SECC); 1197 buildings within 500 m.

Daytime: zoo and park visitors, 11 schools and colleges within 1 km.

Only 4 health places within 1 km: a clinic or nature-education use fits.

Local architectural vocabulary



Front yard

Tiled house, Puttur



Veranda

Pillared front, Melukote



Roof form

Clay-tile roofs, Krishnarajapet



Street edge

Jagali steps, Malavalli

Low, tiled roofs and raised verandas let rural Old Mysore houses sit lightly on sloping, rocky ground: a fit for a park-edge site.

Wikimedia Commons, CC BY-SA 4.0: Prof tpms (Melukote, Krishnarajapet, Malavalli); Bharathesha Alasandemajalu (Puttur) / Old Mysore and coastal Karnataka examples, not on-site records

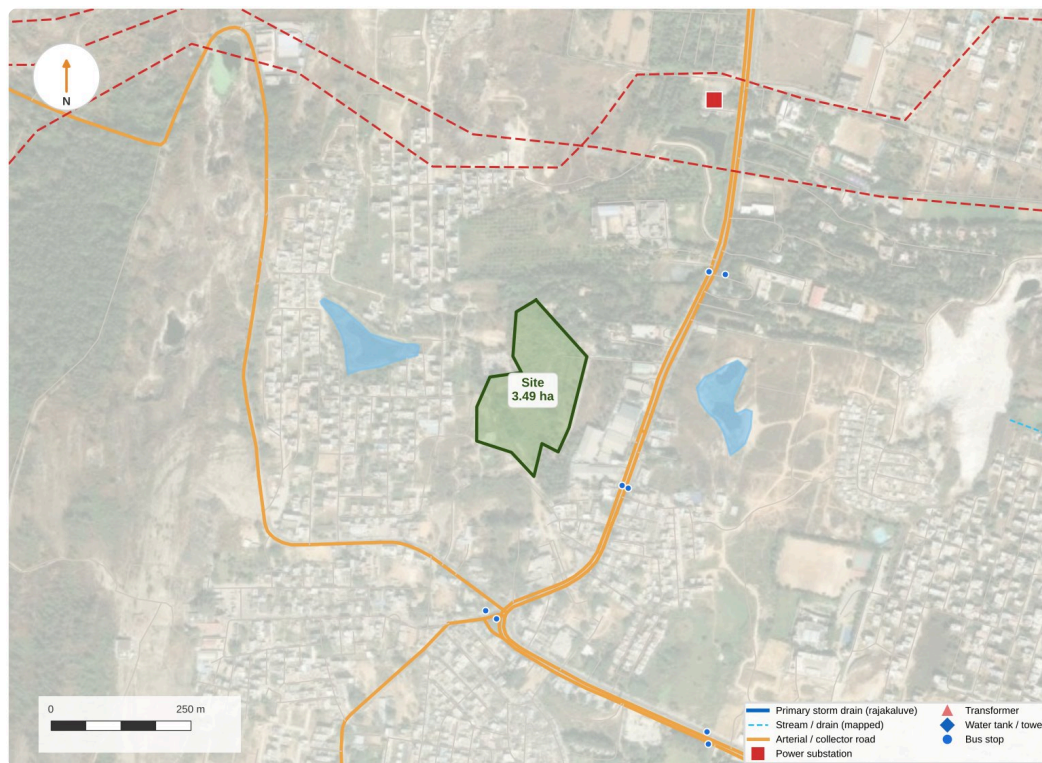
DESIGN / DEVELOPMENT IMPLICATION

Step low, pitched-roof pavilions with deep verandas down the slope, rather than one large footprint: it suits G+1 and the park edge.



Networks, connections & gaps

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru



SOURCE / Overture Maps infrastructure and roads (OpenStreetMap-derived) over Esri imagery. Network capacity needs provider records.

Service register

Service	Connection / distance	Capacity / status	Evidence / next check
Water	No piped main mapped; GP borewells typical	Aquifer over-exploited (Anekal 106%)	GP / BWSSB: supply source
Power	Substation "BG Road" 436 m N; HT line 237 m N	Load not confirmed	BESCOM: feasibility letter
Drainage	Outfall NW corner → village drain → lake 134 m W	No city drain data; slope runoff	GP / BDA: outfall consent
Sewer / waste	No sewer mapped; debris yard on W edge	On-site treatment needed	GP: collection; KSPCB

DESIGN / DEVELOPMENT IMPLICATION

Power is close by, but water, sewer and road access are not. Budget on-site water treatment and reuse, rainwater storage and a STP, and fix the access easement to SH-87 before design.



What matters for this project

Site 1 — Bannerughatta / Bannerughatta, Anekal taluk, Bengaluru

01

Park edge and eco-sensitive zone

Evidence p03, p08 Bannerghatta NP 456 m W. The 2020 ESZ runs 100 m–1 km; the Jan 2026 CEC report proposes restoring the 2016 draft (up to 4.5 km). Commercial resorts and quarrying are barred.

Next: ESZ overlay for Sy 21–22 from the Forest Dept; track the Supreme Court case Owner: client legal Due: before purchase

02

Agricultural zoning

Evidence p03 97.6% Agriculture (draft RMP 2031). Under RMP 2015 AG: farm house $\leq$ 250 m², G+1; other uses need DC conversion.

Next: BDA zonal certificate and PD 30 map Owner: planning consultant Due: concept stage

03

Title and boundary

Evidence p03 Mutation MR 2/2026-27 on Sy 22 awaits a dispute verdict; the Sy 21 point is 63 m outside the KML.

Next: title search; survey sketch vs KML Owner: client surveyor / legal Due: before purchase

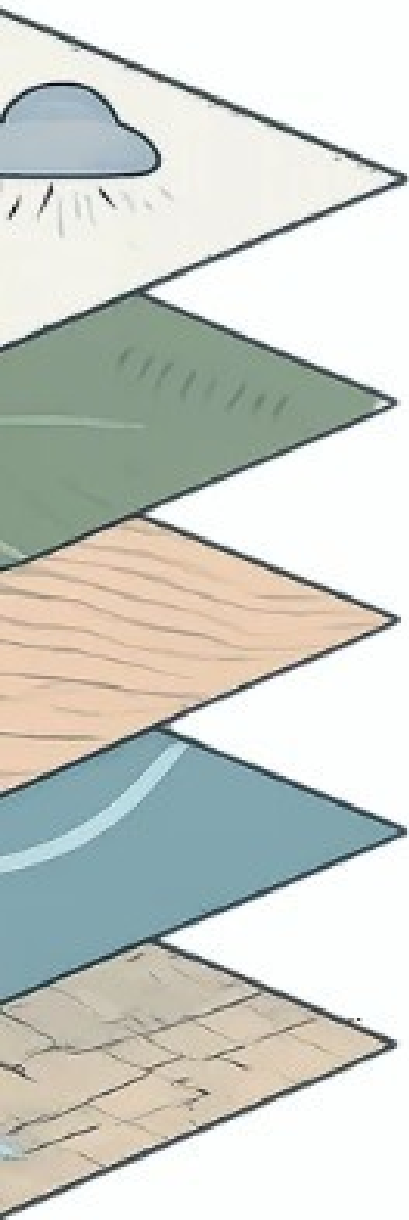
Evidence & coverage

Analysis	Dataset / date	Method / resolution	Status / limitation
Site / controls	KML Oct 2026; RCCMS; RMP 2015 ZR; draft RMP 2031 layer	UTM 43N; plan raster z15 sampling	AG rules from research; PD 30 map not seen
Terrain / water	Copernicus GLO-30; Overture water and park	30 m DSM; D8; Gumbel 1981–2025	DSM includes trees; ESZ line not mapped
Climate / solar	NASA POWER MERRA-2, CERES 2015–24; NOAA ISD HAL	0.5° / 1° grid; station 17.4 km	Regional, not microclimate
Context / services	Overture 2026-09; Esri Wayback 2017 / 2026	Walk 4.8 km/h network; 1.2 km section	Heights untagged; capacity unconfirmed

Not run / unavailable: soil texture (SoilGrids returned 503); ESZ boundary geometry (not in the available datasets); Open-Meteo and OSM Overpass (host blocked, so NASA POWER and Overture used); site borehole and topographic survey.

Prepared by Qnit analysis · GeoAce Studio Reviewed by: pending

Analysis run S01 / 2026-10-10 03:55 UTC Boundary version Case_Studies.kml (Oct 2026)



**Better
design
starts
with
you.**

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